

Pallavi Raguri

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Summary

Detail-oriented and analytical Master's student in Computer Science with strong foundational knowledge in finance, data analysis, and business operations. Experienced in data analytics, customer engagement, and revenue forecasting. Proven ability to leverage tools like Excel, Tableau, and SQL to identify trends and solve problems. Excited to contribute strong communication skills and critical thinking to drive customer satisfaction and improve receivables performance at Norfolk Southern.

Education

MS Computer Science, Clemson University, Clemson, SC, USA | CGPA:3.7 Aug 2023-May 2025

Deep learning, Applied Data Science, Artificial Intelligence, Cloud computing, Statistics, Security in Emerging Systems.

BE Computer Science, Pace Institute of Technology and Sciences, India | CGPA:3.7 Aug 2019 -Jun 2023

Big Data Analytics, Data Structures, Python, OOPS with Java, Database Management System, Operating Systems.

Technical skills

Data Analysis: Python (Pandas, NumPy, Scikit), SQL, Excel

Visualization Tools: Tableau, Power BI, Matplotlib, Seaborn

Machine Learning: Regression, Clustering, Neural Networks, Deep Learning **Cloud**

Platforms: AWS

Analytics: Data Cleaning, Reporting, Trend Analysis, Forecasting

Soft Skills: Customer Service, Communication, Time Management, Problem Solving

Professional Experience

Data Science Intern — SmartKnower Jan 2021 – Jun 2021

- Focused on analyzing and visualizing large-scale datasets using Python, enhancing data integrity and reporting quality.
- Collaborated with cross-functional teams to resolve data discrepancies impacting risk analytics.
- Investigated root causes of data anomalies and recommended corrective actions, improving exposure calculation processes.
- Gained hands-on experience with large-scale data, enhancing analytical thinking and business insight.

Projects

CineBloom – Personalized Movie Recommendation System

- Built a content-based movie recommendation engine using cosine similarity and NLP, achieving 88.5% Precision@10 and 0.869 RMSE.
- Preprocessed TMDB dataset with feature engineering and text normalization to optimize model performance.
- Developed an interactive UI with Streamlit for real-time recommendations and integrated TMDB API for movie posters.
- Proposed future enhancements with collaborative filtering and real-time user feedback to boost personalization.

COVID-19 Forecasting

- Built 30-day COVID-19 case forecasting models for Africa and South America using Python and Facebook Prophet, achieving high predictive accuracy.
- Preprocessed time series data performed feature engineering and tuned model parameters to improve trend prediction performance.
- Developed interactive dashboards using Tableau and Power BI for public health impact.

Machine Learning-Based Health Insurance Price Prediction

- Built machine learning models that predicted insurance premium costs with a 15% improvement in R^2 score after feature engineering and model tuning.
- Stratified customer risk profiles with classification models achieving an F1 score above 0.80.
- Delivered insights that could help optimize real-world insurance pricing strategies and reduce underwriting risks.

Semi-Supervised Machine Learning Approach for DDos Detection

- Developed a hybrid model that detected DDos attacks with over 90% accuracy, while reducing false positives by 25% compared to traditional supervised-only approaches.
- Enhanced model scalability under high-volume network traffic loads, increasing real-time detection speed by 20%.
- Applied semi-supervised learning to maximize performance even on partially labeled datasets.